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MISO DPP 2019 Cycle 1 Phase 1 East (ATC)

System Impact Study Report

J1214, J1244, J1251, J1252, J1253, J1304, J1305, J1316, J1326, J1345, J1370, J1374, J1377, J1410, J1411, J1460, and J1483

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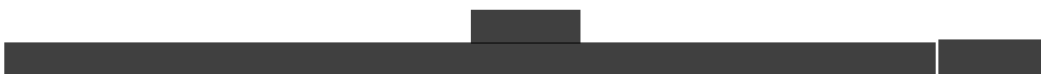


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1.0 EXECUTIVE SUMMARY

Seventeen (17) resource projects have requested to interconnect to the MISO transmission network in the East (ATC) Area and are included in the Definitive Planning Phase 2019 Cycle 1 Phase 1 study (DPP 2019 Cycle 1 Phase 1). All Generating Facilities have requested both Energy Resource Interconnection Service (ERIS) and Network Resource Interconnection Service (NRIS).

This report presents the steady-state study results of a System Impact Study (SIS) performed to evaluate the interconnection of the generator interconnection requests in the DPP 2019 Cycle 1 Phase 1 study. No stability, short circuit or PJM affected system analysis were performed in Phase 1. The study was performed under the direction of Midcontinent Independent System Operator (MISO) by ATC and an ad hoc study group. The results for 2024 scenario are summarized below.

1.1 Project List

The DPP 2019 Cycle 1 Phase 1 has seventeen (17) interconnection requests with a combined nameplate rating of 1,780.08 MW. The DPP 2019 Cycle 1 Phase 1 generator interconnection requests are listed in Table 1.1-1. The one-line diagrams of the interconnection facilities are shown in Appendix C.



Table 1.1-1 – Generating Facilities in DPP April 2018 Phase 1 East (ATC) Area

MISO Queue #	Service Type	Control Area	County, State	Point of Interconnection	Fuel Type	Inverter Model	Main Step-Up Transformer ²	Requested MW	Dispatch (MW) ¹		
									Batteries Discharging		Batteries Charging
									24SH	24SUM	24SH
J1214	NRIS	ALTE	Dane, WI	Rockdale 345 kV	Solar	80 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 336	Two 345/34.5/13.8 kV 105/140/175 MVA Z=9%	300	0	302.4	0
J1244	NRIS	UPPCO	Houghton, MI	Winona – Atlantic 69 kV	Wind	8 x 5.3 MVA Siemens Gamesa 5.0-145 (Type 3) Total Rated MVA = 42.4	Two 69/34.5/13.8 kV 15/20/25 MVA Z=7.5%	38.4	38.4	6.0	38.4
J1251	NRIS	MIUP	Marquette, MI	Perkins – Arnold 138 kV	Solar	28 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 117.6	One 138/34.5/13.8 kV 69/92/115 MVA Z=9%	100	0	100.9	0
J1252	NRIS	MIUP	Marquette, MI	Perkins – Arnold 138 kV	Battery	6 x 4.2 MVA TMEIC ESS-PCS Total Rated MVA = 25.2	One 138/34.5/13.8 kV 15/20/25 MVA Z=9%	20	20.3	20.3	-19.7
J1253	NRIS	WPS	Fond du Lac, WI	South Fond du Lac – Fitzgerald 345 kV	Solar	32 x 3.55 MVA Power Electronics HEM FS3430M Total Rated MVA = 113.6	One 345/34.5/13.8 kV 144/240 MVA Z=9%	100	0	101	0
J1304	NRIS	ALTE	Rock, WI	Townline Road – Paddock 138 kV	Solar	22 x 3.3 MVA Power Electronics Freesun FS3000M Total Rated MVA = 72.6	One 138/34.5/13.8 kV 43/72 MVA Z=9%	65	0	65.8	0
J1305	NRIS	ALTE	Green, WI	Bass Creek – Albany – North Monroe 138 kV	Solar	17 x 3.3 MVA Power Electronics Freesun FS3000M Total Rated MVA = 56.1	One 138/34.5/13.8 kV 33/55 MVA Z=9%	50	0	50.6	0
J1316	NRIS	WEC	Kenosha, WI	Paris 138 kV	Battery	14 x 4.2 MVA TMEIC ESS 840 Total Rated MVA = 58.8	One 138/34.5/13.8 kV 36/48/60 MVA Z=9%	50	50.6	50.6	-49.4
J1326	NRIS	ALTE	Dane, WI	Rockdale 345 kV	Battery	20 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 84	One 345/34.5/13.8 kV 54/72/90 MVA Z=9%	75	75.6	75.6	-74.1
J1345	NRIS	ALTE	Dodge, WI	North Beaver Dam 69 kV	Battery	9 x 3.5 MVA Power Electronics HEC-US FS3000CU15 Total Rated MVA = 34.2	One 69/34.5/13.8 kV 51/85 MVA Z=8.5%	25	25.2	25	-24.8
J1370	NRIS	MIUP	Chippewa, MI	Dafter 69 kV	Gas	N/A	One 69/12.47 kV 40/66.8 MVA Z=8%	50	0	50	0
J1374	NRIS	ALTE	Grant, WI	Ebenezer 138 kV	Wind	17 x 6.2 MVA Vestas V162-5.6 (Type 4) Total Rated MVA = 105.4	One 138/34.5/13.8 kV 66/88/110 MVA Z=10.5%	95	95.2	14.85	95.2
J1377	NRIS	ALTE	Rock, WI	Rockdale – Wempletown 345 kV	Solar	32 x 3.46 MVA Power Electronics HEM FS3350M Total Rated MVA = 110.72	One 345/34.5/13.2 kV 69/92/115 MVA Z=7%	98.56	0	98.56	0
J1410	NRIS	MGE	Dane, WI	North Madison 345 kV	Solar	80 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 336	Two 345/34.5/13.8 kV 105/140/175 MVA Z=9%	300	0	302.4	0

J1411	NRIS	MGE	Dane, WI	North Madison 345 kV	Battery	20 x 4.2 MVA TMEIC PVU-L0840GR Total Rated MVA = 84	One 138/34.5/13.8 kV 54/72/90 MVA Z=9%	75	75.6	75.6	-74.1
J1460	NRIS	ALTE	Rock, WI	Sunrise 138 kV	Solar	60 x 3.59 MVA Power Electronics HEMK 690V FS3450K Total Rated MVA = 215.4	Two 138/34.5/13.8 kV 75/100/125 MVA Z=9%	200	0	202.4	0
J1483	NRIS	ALTE	Iowa, WI	Hill Valley 345 kV	Wind	44 x 2.653 MVA GE 2.5-127-60 Hz (Type 3) Total Rated MVA = 116.732	One 345/34.5/13.8 kV 72/96/120 MVA Z=9%	99	100.7	15.44	100.7

¹ Per MISO BPM 015-r22 the following dispatch assumptions are applied for each Fuel Type.

- a. Combined Cycle (CC) is dispatched to 50% of the Requested MW in the shoulder models, 100% in the summer peak model.
- b. Solar is dispatched offline in the shoulder models, 100% in the summer peak model.
- c. Wind is dispatched to 100% of the Requested MW in the shoulder models, 15.6% in the summer peak model.
- d. Battery is dispatched to 100% of the Requested MW in discharging mode in the summer peak model and dispatched in both charging and discharging modes to +/-100% of the Requested MW in the shoulder models.
- e. Hybrid is dispatched according to Appendix E of MISO BPM 015-r22.

² Three winding transformers add a node not assigned to a PSSE bus number which must be initialized for every model build. Past experience has shown that these nodes not assigned to a PSSE bus number have been a source for power flow model solution difficulties. Since the third winding, when it has no devices such as a shunt or load connected to it, has no impact on the steady-state modeling ATC has opted to not model them. The three-winding transformers will be modeled as a three-winding transformer for the short circuit study



1.2 Generating Facility Requirements

1.2.1 Voltage Schedule Requirement

ATC requires all generators in its territory to maintain a voltage schedule at the Point of Interconnection (POI). The standard voltage schedule is 1.02 per unit as measured at the POI. This schedule may be changed by the Transmission Operator for specific power plants or specific conditions.

1.2.2 Power Factor Range Requirement

FERC Order 827 and ATC Criteria require all newly interconnecting generators interconnecting to ATC-owned Facilities to provide a power factor range for synchronous and non-synchronous (e.g., wind turbines, solar) generation of 0.95 leading (when a Generating Facility is consuming reactive power from the Transmission System) to 0.95 lagging (when a Generating Facility is supplying reactive power to the Transmission System). The Generating Facility must be capable of maintaining ATC's standard power factor range at all power output levels by providing dynamic reactive power at the following locations:

- A. The POI for all synchronous generators
- B. The high-side of the generator substation for all non-synchronous generators

For synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the POI. For non-synchronous machines, the interconnection studies will account for the net effect of all energy production devices and losses on the Customer's side of the generator substation. Dynamic reactive power provided by non-synchronous generators must meet the following requirement from FERC order 827 Item 35:

"Non-synchronous generators may meet the dynamic reactive power requirement by utilizing a combination of the inherent dynamic reactive power capability of the inverter, dynamic reactive power devices (e.g., Static VAR Compensators), and static reactive power devices (e.g., capacitors) to make up for losses."

Therefore, static reactive power sources can only be used to make up for losses between the terminal of the machines and the high side of the generator substation for non-synchronous machines. All other reactive power to meet the power factor requirement must be provided by dynamic sources. Static sources can be switched on or off in the range of seconds and provide reactive power in large discrete blocks. Capacitor Banks are considered static sources of reactive power. Dynamic sources can provide variable amounts of reactive power in a few milliseconds. Static Var Compensators (SVCs), Static Synchronous Compensators (STATCOMs), Flexible AC Transmission Systems (FACTS), inverters and synchronous condensers are all considered dynamic sources of reactive power.

For non-synchronous generation projects in the DPP 2019 Cycle 1 Phase 1 study group, if they did not have a signed Generator Interconnection Agreement (GIA) or Provisional GIA (PGIA) by September 21, 2016, they are required to provide dynamic reactive power within the range of 0.95 leading to 0.95 lagging at the high-side of the generator substation.

The generation requests shown in Table 1.2.2-1 did not meet the dynamic reactive power requirements per FERC Order 827 and ATC Criteria at the time of the model completion and are required to provide additional dynamic reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for dynamic reactive power requirements.

Table 1.2.2-1 – Additional Dynamic Mvar to meet ATC Dynamic Inductive Power Factor Requirement and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Dynamic Reactive Compensation (Mvar)
J1460	Asynchronous	-8.21

The generation requests shown in Table 1.2.2-2 did not meet the static reactive power requirements per ATC Criteria at the time of the model completion and are required to provide additional static reactive power sources. All other requests in this queue met FERC Order 827 and ATC Criteria for static reactive power requirements.

Table 1.2.2-2 – Additional Static Mvar to meet ATC Capacitive Power Factor and FERC Order 827 Power Factor Requirements

MISO Queue #	Type	Additional Static Shunt Compensation ¹ (Mvar)
J1251/J1252	Asynchronous	4.4
J1304	Asynchronous	6.4
J1305	Asynchronous	4.9
J1483	Asynchronous	18.2

¹ Additional compensation is required to meet the criteria at the POI Bus for synchronous Generating Facilities or the high-side of the generator substation for asynchronous Generating Facilities.

The generation requests shown in Table 1.2.2-3 may need an assessment of power factor at low real power output levels. More information regarding reactive power capability under low real power output level is required from the interconnection customers to determine whether the study is needed. All other requests in this queue met FERC Order 827 and ATC Criteria and do not need an assessment of power factor requirements at low real power output levels.

Table 1.2.2-3 – Assessment of Power Factor Requirements at Low Output Levels

MISO Queue #	P-Q Curve Type	Is Power Factor Evaluation Needed for Minimum Output Levels?	Additional Static Mvar required at POI Bus (synchronous) or HV Bus (asynchronous) to meet ATC Capacitive Power Factor Requirement
J1214/J1326	D-shape	No	N/A
J1244	U-Shape	Yes	0.8
J1251/J1252	D-shape	No	N/A
J1253	D-shape	No	N/A
J1304	D-shape	No	N/A
J1305	D-shape	No	N/A
J878/J1316	D-shape	No	N/A
J1003/J1345	D-shape	No	N/A
J1370	D-shape	No	N/A
J1374	D-shape	No	N/A
J1377	D-shape	No	N/A
J1410/J1411	D-shape	No	N/A
J1460	Rectangular-shape	No	N/A
J1483	Rectangular-shape (reduced at Min output levels)	Yes	No

1.2.3 Island Detection and Operation

In circumstances where the Generating Facility has no governor controls and the transmission system design could result in an islanding condition for the outage of two transmission elements, ATC requires the Customer to implement additional protection systems as mutually agreed by the Customer and ATC to prevent generation from being isolated or islanded with interconnected load. Alternatively, ATC will require the Customer to curtail their generation for circumstances that could result in an island condition with the next contingency.

This would apply to the following Generating Facilities from this DPP cycle that lack adequate governor controls to safely and reliably sustain an island with load.

- J1305
- J1374

1.3 Total Network Upgrades for all Projects

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Energy Resource Interconnection Service and Network Resource Interconnection Service as of the SIS report date. The total cost of Network Upgrades required for each generator interconnection request is listed in Table 1.3-1. The costs for Network Upgrades are planning level estimates and subject to

be revised in the facility studies. No Network Upgrade projects driven by stability, short circuit and PJM affected system analysis were not identified since these analyses were not performed in Phase 1. All Interconnection Facility Project Diagrams are documented in Appendix C and all Network Upgrade Project Diagrams are documented in Appendix D (No project diagrams are developed for line uprates).



Table 1.3-1 – Total Cost of Network Upgrades for DPP Cycle 1 Phase 1 Generator Interconnection Requests

MISO Queue #	Requested MW	ERIS Network Upgrades (\$) ⁷				NRIS Network Upgrades (\$)	Interconnection Facilities (\$)		Shared Network Upgrade (\$)	Total Cost of Network Upgrades (Exclude TOIF & Affected Systems) (\$)	M2 Received (\$) ¹	M3 Due (\$) ²
		Steady - State Thermal & Voltage	Transient Stability	Short Circuit	Affected System		TO Network Upgrades	TO-Owned Direct Assigned (TOIF)				
[a]	[b]	[c]	[d]	[e]	[f]	[g]	[h]	[i]	[j]	[k]=[c]+[d]+[e]+[g]+[h]	[l] = \$4000 x [b]	[m]= (10%of [k])-[l]
J1214	300	0	0	0	0	6,369,605	859,833	256,232	0	7,229,438	1,200,000	0
J1244	38.4	0	0	0	0	584,079	6,385,560	260,000	0	6,969,639	153,600	543,364
J1251	100	0	0	0	0	0	3,956,000	145,000	0	3,956,000	400,000	0
J1252	20	0	0	0	0	0	3,956,000	145,000	0	3,956,000	80,000	315,600
J1253	100	0	0	0	0	4,666,742	13,498,520	467,480	0	18,165,262	400,000	1,416,526
J1304	65	0	0	0	0	4,665,837	7,836,034	273,966	0	12,501,871	260,000	990,187
J1305	50	0	0	0	0	3,771,830	7,998,040	281,960	0	11,769,870	200,000	976,987
J1316	50	0	0	0	0	292,629	0	0	0	292,629	200,000	0
J1326	75	0	0	0	0	1,592,415	859,833	256,232	0	2,452,248	300,000	0
J1345	25	0	0	0	0	20,096,868	0	0	0	20,096,868	100,000	1,909,687
J1370	50	325,098	0	0	0	25,477,000	0	0	0	25,802,098	200,000	2,380,210
J1374	95	0	0	0	0	5,955,736	767,975	255,275	0	6,723,711	380,000	292,371
J1377	98.56	0	0	0	0	1,991,351	11,966,628	580,000	0	13,957,979	394,240	1,001,558
J1410	300	0	0	0	0	9,287,928	5,336,386	392,381	0	14,624,313	1,200,000	262,431
J1411	75	0	0	0	0	2,323,507	5,336,386	392,381	0	7,659,892	300,000	465,989
J1460	200	0	0	0	0	29,070,959	4,881,028	337,972	0	33,951,987	800,000	2,595,199
J1483	99	3,518,467	0	0	0	4,726,286	1,447,638	471,921	0	9,692,391	396,000	573,239
Total (\$)	-	3,843,565	0	0	0	120,872,772	75,085,860	4,515,800	0	199,802,197	6,963,840	13,723,348

¹ M2: Milestone Payment dollars received by MISO
² M3 = (10% of NU)-M2
³ ATC projects are estimated in ISD dollars with a 2.5% annual escalation rate and include a 10% contingency.
⁴ Pre-Certification costs are not included.
⁵ Transmission Owner shall also collect from Interconnection Customer a tax gross-up amount on the payments made to Transmission Owner using the Transmission Owner rate in effect at the time the payment is received from Interconnection Customer. The current Transmission Owner tax gross-up rate is 12.848%.
⁶ TOIF: Non-reimbursable Transmission Owner Interconnection Facilities to which tax gross-up amount must also be applied.
⁷ Transient stability, short circuit, and affected system studies will be performed as part of MISO DPP 2019 Cycle 1 Phase 2.

1.4 In-Service Dates and Cost Estimates

ATC understands that the estimated in-service date may not align with the Interconnection Customer's Synchronization Date; however, negotiated and executed agreements, such as an Engineering and Procurement Agreement, can be used prior to the GIA execution date to expedite Network Upgrades. In absence of any special arrangement, typical times to develop a new Interconnection Facility is about 24-36 months after the GIA is executed, assuming no delays due to Interconnection Customer's permits, state processes, land acquisitions, deliverables (such as a finish graded substation site, etc.) It also assumes that system outages required to construct facilities can be obtained timely. The cost estimates for Interconnection Facilities are based on the in-service date provided in the Interconnection Customer's application data. Therefore, any change in in-service date will have impact on the cost estimates. The requested dates for Interconnection Facility in-service, synchronization, and commercial operation are summarized in Table 1.4-1.

Table 1.4-1 – Requested Interconnection Facilities In-Service Dates, Synchronization Dates and Commercial Operation Dates

MISO Queue #	Requested Interconnection Facility In-service Date	Requested Synchronization Date	Requested Commercial Operation Date
J1214	October 30, 2022	October 30, 2022	December 31, 2022
J1244	September 15, 2021	October 5, 2021	October 29, 2021
J1251	October 30, 2022	October 01, 2022	December 31, 2022
J1252	October 30, 2022	October 01, 2022	December 31, 2022
J1253	August 1, 2022	September 1, 2022	December 31, 2022
J1304	December 1, 2021	January 2, 2022	May 1, 2022
J1305	December 1, 2021	January 2, 2022	May 1, 2022
J1316	October 30, 2022	October 30, 2022	December 31, 2022
J1326	October 30, 2022	October 30, 2022	December 31, 2022
J1345	October 1, 2021	October 15, 2021	December 1, 2021
J1370	March 1, 2021	April 1, 2021	June 1, 2021
J1374	August 1, 2021	October 1, 2021	December 1, 2021
J1377	July 15, 2022	August 15, 2022	October 31, 2022
J1410	October 30, 2022	October 30, 2022	December 31, 2022
J1411	October 30, 2022	October 30, 2022	December 31, 2022
J1460	June 30, 2022	August 31, 2022	October 31, 2022
J1483	October 30, 2023	October 1, 2023	December 31, 2023

1.5 MTEP Projects

If a MTEP transmission project(s) resolves the constraint, and that project(s) is approved by the Board within (1) calendar year of the GIA execution or execution of an amendment thereof, then the Interconnection Customer will not be responsible for transmission upgrade(s) that would resolve the constraint. If that MTEP project(s) is not approved within one (1) calendar year of the GIA execution or execution of an amendment thereof, the Interconnection Customer will be responsible for those transmission upgrade(s).

1.6 Further Study

The next step in the MISO Generator Interconnection Procedures is to perform additional SISs (if needed), Interconnection Customer Interconnection Facility Studies, and Network Upgrade Facility Studies. Those Facilities Studies will specify in more detail the time and cost of the equipment, engineering, procurement, and construction of the Interconnection Facilities and Network Upgrades identified in this report.

1.7 Compliance Summary

This study report partially meets NERC TPL-001-4 standard, FAC-002-2 standard, and Local Planning Criteria. In ATC's annual Ten-Year Assessment (TYA) and MISO annual MTEP studies, additional compliance related studies will be performed for the generator interconnection requests with signed GIAs. Appendix J describes in detail the NERC and Local Criteria requirements met by this SIS report.

2.0 STEADY-STATE ANALYSIS

Steady-state analysis was performed to identify thermal and voltage upgrades required to interconnect the generator interconnection requests in the DPP 2019 Cycle 1 Phase 1 to the transmission system. Detailed study assumptions, criteria, and methodology are documented in Appendix A.

2.1 Model Development

2.1.1 Study Cases

Four study cases for the steady-state thermal and voltage analysis were developed based upon the expected topology for the local area: 2024 summer peak batteries discharging, 2024 summer peak batteries charging, 2024 shoulder batteries discharging, and 2024 shoulder batteries charging. The ATC system in these cases was updated with the most recent data available at the time of model construction. The cases were reviewed by ATC and the Interconnection Customers. Based on this review, the cases were further modified to account for model updates, changes, and competing generation requests that had dropped out of the MISO queue since the models were built.

The prior queued generator interconnection requests in the ATC system that are included in the study cases are listed in Table 2.1.1-1. Associated Network Upgrades were also included based on their expected in-service date.

**Table 2.1.1-1 – Prior Queued Generator Interconnection Requests
Not Yet In-Service**

MISO Queue #	Type	Control Area	Requested MW
J505	Solar	WPS	99
J584	Wind	ALTE	60
J732	CC	WPS	561.5
J818	Solar	WEC	149
J849	Solar	MIUP	125
J850	Solar	ALTE	250
J855	Wind	ALTE	100
J864	Solar	ALTE	49.98
J870	Solar	ALTE	200
J871	Solar	ALTE	100
J878	Solar	WEC	200
J886	Solar	WPS	150
J928	Wind	MIUP	79.995
J947	Solar	ALTE	200
J986	Solar	ALTE	149.76
J1000	Solar	ALTE	50
J1002	Solar	ALTE	99
J1003	Solar	ALTE	50
J1042	Solar	ALTE	180
J1101	Battery	WPS	20
J1153	Solar	WEC	150
J1154	Solar	WEC	75
J1171	Solar	WEC	100
J1183	Solar	MIUP	1.35
J1188	Solar	ALTE	50

Public information related to the MISO Generator Interconnection Request queue can be found at:

https://www.misoenergy.org/planning/generator-interconnection/GI_Queue/

The summer peak discharging, summer peak charging, shoulder discharging, and shoulder charging cases dispatched generation within MISO according to section 6.1.1.1.2, Study Case Development, in the MISO BPM-015-r22.

All excess generation from this methodology is dispatched against all units in MISO Classic proportionally, excluding the units in the current DPP cycle. Scheduled firm transfers are ignored in this dispatch methodology.

2.1.2 Benchmark Cases

Two benchmark cases were used to benchmark system performance without the DPP 2019 Cycle 1 Phase 1 generating facilities and were created by taking the DPP 2019 Cycle 1 Phase 1 Generating Facilities offline from the corresponding study cases. The MISO Classic was used for power balance, where generation was scaled in proportion to Pmax minus Pgen.

2.2 Reactive Power Requirements (FERC Order 827)

All synchronous and non-synchronous generation in this queue were evaluated to determine if the requests meet FERC Order 827 and ATC Planning Criteria. Refer to PLG-METH-0005 in Appendix B for details on the methodology used to determine power factor compliance. All of the reactive resources modeled in the assessment are summarized in Table 2.2-1.

Table 2.2-1 – Reactive Resources Modeled in Generator Interconnection Power Factor Analysis

MISO Queue #	Machines				Dynamic Devices			Static Devices		
	Description	Real Power (MW)	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)	Description	Capacitive Reactive Power (Mvar)	Inductive Reactive Power (Mvar)
J1214/J1326	Solar Machines/Battery Storage	378	183.07	-183.07	None	N/A	N/A	2-4 MVAR Cap Bank	8	0
J1244	Wind Turbines	38.4	19.4	-19.4	None	N/A	N/A	None	0	0
J1251/J1252	Solar Machines/Battery Storage	128.52	62.24	-62.24	None	N/A	N/A	1-3 MVAR Cap Bank	3	0
J1253	Solar Machine	101	52	-52	None	N/A	N/A	None	0	0
J1304	Solar Machine	66	30.24	-30.24	None	N/A	N/A	None	0	0
J1305	Solar Machine	51	23.37	-23.37	None	N/A	N/A	None	0	0
J878/J1316	Solar Machines/Battery Storage	252.92	141.09	-141.09	None	N/A	N/A	1-2 MVAR Cap Bank	2	0
J1003/J1345	Solar Machines/Battery Storage	77	45.89	-45.89	None	N/A	N/A	None	0	0
J1370	Simple Cycle	50	28	-24.68	None	N/A	N/A	None	0	0
J1374	Wind Turbines	95.2	45.288	-33.779	None	N/A	N/A	2-6 MVAR Cap Bank	12	0
J1377	Solar Machine	98.56	50.494	-50.494	None	N/A	N/A	None	0	0
J1410/J1411	Solar Machines/Battery Storage	378	183.07	-183.07	None	N/A	N/A	2-4 MVAR Cap Bank	8	0
J1460	Solar Machines	204.6	67.43	-59.05	None	N/A	N/A	2-26 MVAR Cap Bank	52	0
J1483	Wind Turbines	110.88	36.44	-36.44	None	N/A	N/A	1-3 MVAR Cap Bank	3	0

The dynamic capacitive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-2.

Table 2.2-2 – Assessment of Dynamic Capacitive Power Factor Requirement

MISO Queue #	Machine Terminal Bus #	Capability at Machine Terminal		Additional Dynamic Capacitive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power (MW)	Capacitive Reactive Power (Mvar)				
J1214/J1326	43260, 43261, 43262	378.0	183.000	0.0	0.90	Yes	0.0
J1244	42440, 42441	38.4	19.400	0.0	0.89	Yes	0.0
J1251/J1252	42520, 42521	128.5	62.300	0.0	0.90	Yes	0.0
J1253	42530	101.0	52.000	0.0	0.89	Yes	0.0
J1304	43040	65.8	30.200	0.0	0.91	Yes	0.0
J1305	43054	50.6	23.400	0.0	0.91	Yes	0.0
J878/J1316	88784, 88784, 43160	252.9	141.000	0.0	0.87	Yes	0.0
J1003/J1345	40030, 43450	77.0	45.900	0.0	0.86	Yes	0.0
J1370	43700	50.0	28.000	0.0	0.87	Yes	0.0
J1374	43740	95.2	45.288	0.0	0.90	Yes	0.0
J1377	43770	98.56	50.494	0.0	0.89	Yes	0.0
J1410/J1411	44110, 44111, 44112	378.0	183.000	0.0	0.90	Yes	0.0
J1460	44600, 44601	204.6	67.430	0.0	0.95	Yes	0.0
J1483	44830	110.88	36.440	0.0	0.95	Yes	0.0

¹ Dynamic capacitive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Capacitive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Capacitive Dynamic Power Factor.

The static capacitive power factor requirement analysis showed 4 of the 17 requests do not meet the ATC Criteria or FERC Order 827 requirements. The results are summarized in Table 2.2-3.

Table 2.2-3 – Assessment of Static Capacitive Power Factor Requirement

MISO Queue #	Point of Measurement	Capability at Point of Measurement ¹		Power Factor	Meets Requirement?	Additional Requirement (Mvar)
		Real Power (MW)	Reactive Power (Mvar)			
J1214/J1326	43269	371.5	158.8	0.92	Yes	0.0
J1244	42446	37.6	12.7	0.95	Yes	0.0
J1251/J1252	42526	127.1	37.4	0.96	No	4.4
J1253	42534	100	35.8	0.94	Yes	0.0
J1304	43043	64.9	14.9	0.97	No	6.4
J1305	43053	49.8	11.5	0.97	No	4.9
J878/J1316	88780	248.9	90.9	0.94	Yes	0.0
J1003/J1345	40033	75.5	30.7	0.93	Yes	0.0
J1370	43703	49.5	19.6	0.93	Yes	0.0
J1374	43743	92.76	34.47	0.94	Yes	0.0
J1377	43773	96.7	32.25	0.95	Yes	0.0
J1410/J1411	44119	371.7	167.6	0.91	Yes	0.0
J1460	44605	202.2	80.85	0.93	Yes	0.0
J1483	44833	108.64	17.52	0.99	No	18.2

¹ Point of Measurement is the POI Bus for synchronous machines and high side of generator substation for asynchronous machines.

The dynamic inductive power factor requirement analysis showed all but one of the requests meet ATC Criteria and FERC Order 827 requirements. J1460 does not meet the ATC Criteria and FERC Order 827 requirements and must provide additional dynamic resources prior to Phase two. The results are summarized in Table 2.2-4.

Table 2.2-4 – Assessment of Dynamic Inductive Power Factor Requirement

MISO Queue #	Machine Terminal Bus #	Capability at Machine Terminal		Additional Dynamic Inductive Reactive Power ¹ (Mvar)	Dynamic Power Factor Provided	Meets Requirement ² ?	Additional Requirement ³ (Mvar)
		Real Power (MW)	Inductive Reactive Power (Mvar)				
J1214/J1326	43260, 43261, 43262	378.0	-183	0.0	0.90	Yes	0.0
J1244	42440, 42441	38.4	-19.4	0.0	0.89	Yes	0.0
J1251/J1252	42520, 42521	128.5	-62.3	0.0	0.90	Yes	0.0
J1253	42530	101.0	-52	0.0	0.89	Yes	0.0
J1304	43700	65.8	-30.2	0.0	0.91	Yes	0.0
J1305	43700	50.6	-23.4	0.0	0.91	Yes	0.0
J878/J1316	88784, 88784, 43160	252.9	-141	0.0	0.87	Yes	0.0
J1003/J1345	40030, 43450	77.0	-45.9	0.0	0.86	Yes	0.0
J1370	43700	50.0	-24.7	0.0	0.90	Yes	0.0
J1374	43740	95.2	-33.779	0.0	0.94	Yes	0.0
J1377	43770	98.56	-50.494	0.0	0.89	Yes	0.0
J1410/J1411	44110, 44111, 44112	378.0	-183	0.0	0.90	Yes	0.0
J1460	44600, 44601	204.6	-59.05	0.0	0.96	No	-8.21
J1483	44830	110.88	-36.44	0.0	0.95	Yes	0.0

¹ Dynamic inductive reactive power provided by Interconnection Customer owned equipment in addition to the machine.

² ATC requires a 0.95 ATC Inductive Dynamic Power Factor.

³ Additional dynamic reactive power required to meet ATC Inductive Dynamic Power Factor.

The static inductive power factor requirement analysis showed all requests meeting ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-5.

Table 2.2-5 – Assessment of Static Inductive Power Factor Requirement

MISO Queue #	POI Bus (synchronous) or HV Bus (asynchronous)	MW at POI (synchronous) or HV Bus (asynchronous)	Mvar at POI (synchronous) or HV Bus (asynchronous)	POI (synchronous) or HV Bus (asynchronous) Power Factor	Meets 0.95 ATC Inductive Power Factor Requirement at POI Bus (synchronous) or HV Bus (asynchronous)?
J1214/J1326	43269	370.9	-219.1	0.86	Yes
J1244	42446	37.5	-28	0.80	Yes
J1251/J1252	42526	126.9	-95.5	0.80	Yes
J1253	42534	99.8	-70.3	0.82	Yes
J1304	43043	64.6	-49.1	0.80	Yes
J1305	43053	49.6	-38	0.79	Yes
J878/J1316	88780	248.0	-205.7	0.77	Yes
J1003/J1345	40033	74.5	-75.9	0.70	Yes
J1370	43703	49.5	-34.4	0.82	Yes
J1374	43743	92.5	-58.83	0.84	Yes
J1377	43773	96.4	-72.17	0.80	Yes
J1410/J1411	44119	371.3	-209	0.87	Yes
J1460	44605	202.3	-94.69	0.91	Yes
J1483	44833	108.5	-59.42	0.88	Yes

The low output power factor requirement analysis showed all requests meet ATC Criteria and FERC Order 827 requirements. The results are summarized in Table 2.2-6.

Table 2.2-6 – Assessment of Power Factor Requirements at Low Output Levels

MISO Queue #	P-Q Curve Type	Is Power Factor Evaluation Needed for Minimum Output Levels?	Additional Static Mvar required at POI Bus (synchronous) or HV Bus (asynchronous) to meet ATC Capacitive Power Factor Requirement
J1214/J1326	D-shape	No	N/A
J1244	U-shape	Yes	0.8
J1251/J1252	D-shape	No	N/A
J1253	D-shape	No	N/A
J1304	D-shape	No	N/A
J1305	D-shape	No	N/A
J878/J1316	D-shape	No	N/A
J1003/J1345	D-shape	No	N/A
J1370	D-shape	No	N/A
J1374	D-shape	No	N/A
J1377	D-shape	No	N/A
J1410/J1411	D-shape	No	N/A
J1460	D-shape	No	N/A
J1483	Rectangular	Yes	No

2.3 NERC TPL Contingency Analysis Results

The incremental impact of the proposed generator interconnection on transmission facilities was evaluated by comparing steady state power flows and voltages between benchmark cases (without DPP 2019 Cycle 1 Phase 1 projects) and study cases (with DPP 2019 Cycle 1 Phase 1 projects). Post-contingency cases were solved with transformer tap adjustment enabled, area interchange adjustment disabled, phase shifter adjustment enabled, and switched shunt adjustment enabled. Detailed NERC TPL Category P contingencies that were studied are described in Table A.2.1-1 in Appendix A.

2.3.1 2024 Summer Peak Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2024 Summer Peak Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2024 Summer Peak Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.1. No voltage constraints were identified in the steady-state analyses.

Table 2.3.1 – 2024 Summer Peak Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2024 Summer Discharging	Dafter – Nine Mile 69 kV line	ATC	Base Case P11 P12 P21	J1370

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.2 2024 Shoulder Batteries Discharging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2024 Shoulder Discharging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2024 Shoulder Discharging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2. No voltage constraints were identified in the steady state analyses.

Table 2.3.2 – 2024 Shoulder Batteries Discharging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2024 Shoulder Discharging	J1244 POI – Portage Tap – Atlantic 69 kV line	ATC	Base Case P12 P13 P14 P21 P23	J1244
	J1244 POI – Winona 69 kV line	ATC	P12 P21	J1244
	Highland – Eden 138 kV line	ATC	P12	J1483

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.3 2024 Shoulder Batteries Charging

The study identified the steady-state thermal and voltage constraints that qualified as MISO Injection Constraints in the 2024 Shoulder Charging study model under NERC Category P0-P7 Planning Events (except NERC Category P3 and P6). Detailed steady-state power flow results and Injection Constraint determination can be found in Appendix E. A summary of the 2024 Shoulder Charging MISO Injection Constraints that require Network Upgrades is presented in Table 2.3.2-4. No voltage constraints were identified in the steady state analyses.

Table 2.3.4 – 2024 Shoulder Batteries Charging Steady-State Injection Constraints Requiring Network Upgrades

Study Case	Overloaded Facility	Facility Owner	Resulted by "No Load Loss Allowed" ¹ NERC TPL Planning Events	Responsible Generator(s)
2024 Shoulder Charging	J1244 POI – Portage Tap – Atlantic 69 kV line	ATC	Base Case P11 P12 P13 P14 P21 P23	J1244
	J1244 POI – Winona 69 kV line	ATC	P12 P21	J1244
	Highland – Eden 138 kV line	ATC	P12	J1483

¹ The "No Load Loss Allowed" NERC TPL Planning Events refer to all the Planning Events in NERC TPL-001-4 Table 1 that the interruption of Firm Transmission Service and Non-Consequential Load Loss are not allowed.

² Per ATC's Planning Criteria, for Planning Events that allow Non-Consequential load loss, short duration exposure to potential post-contingency overload represents an acceptable risk assuming system adjustment within that short period of time can eliminate the exceedance and restore facility loading within Long-Term emergency ratings.

2.3.4 Network Upgrades Identified in ERIS Analysis

Based on the steady-state analyses, the worst loading of each facility under "No Load Loss Allowed" NERC TPL Planning Events that meets MISO Injection Constraint criteria is shown in Table 2.3.3-1. Potential Network Upgrades are also included. Good faith Cost Estimates of the ERIS thermal Network Upgrades

identified in the steady-state analysis for the 2024 scenarios are listed in Table 2.3.3-2. Detailed cost allocations are provided in Section 9.

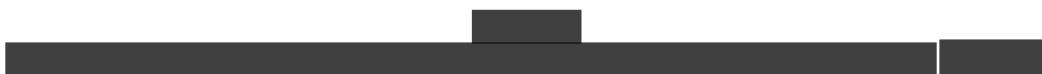


Table 2.3.5-1 – ERIIS Network Upgrades Identified to Address
MISO Steady-State Injection Constraints

Steady-State Injection Constraint	Responsible Generator(s)	Facility Owner	Study Case	Applicable Rating (MVA)	Worst Loading (%)	"No Load Loss Allowed" NERC TPL Planning Event	Network Upgrades
Dafter – Nine Mile 69 kV line	J1370	ATC	24SUM, Batteries Discharging	24	217.79		Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate
J1244 POI – Portage Tap 69 kV line	J1244	ATC	24SH, Batteries Discharging	37	110.18		Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP PrjId#4727
Portage Tap – Atlantic 69 kV line	J1244	ATC	24SH, Batteries Discharging	37	110.10		
J1244 POI – Winona 69 kV line	J1244	ATC	24SH, Batteries Charging	37	107.19		
Highland – Eden 138 kV line	J1483	ATC	24SH, Batteries Charging	309	106.53		Highland – Eden (X-147) 138 kV, Rebuild

Table 2.3.5-2 – ERIS Network Upgrades and Cost Estimates

Steady-State Injection Constraint	Facility Owner	Network Upgrade	Cost (\$) ^{1,2}
Dafter – Nine Mile 69 kV line	ATC	Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate	325,098
J1244 POI – Portage Tap 69 kV line	ATC	Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP ProjId#4727	0
Portage Tap – Atlantic 69 kV line	ATC		
J1244 POI – Winona 69 kV line	ATC		
Highland – Eden 138 kV line	ATC	Highland – Eden (X-147) 138 kV, Rebuild	3,518,467

¹ All Network Upgrades were estimated on the generator ISD dollars.

² ATC Network Upgrades included a 10% contingency.

2.3.5 Network Upgrade Alternatives Considered

The ERIS network upgrades identified in Table 2.3.3-2 are direct upgrades of the ERIS thermal constraint facilities to ATC design standards and considered as least-cost solutions. Therefore, no other alternatives were examined for those solutions.

2.3.6 Potential Operating Restriction

Operating restriction analysis will be performed as part of MISO DPP 2019 Cycle 1 Phase 2.

3.0 STABILITY ANALYSIS

Stability analysis will be performed as part of MISO DPP 2019 Cycle 1 Phase 2.

4.0 SHORT CIRCUIT ANALYSIS

Short circuit analysis will be performed as part of MISO DPP 2019 Cycle 1 Phase 2.

5.0 TRANSFORMER ENERGIZATION ANALYSIS

5.1 Transformer Energization Study Results

Transformer Energization Analysis were performed based on ATC transformer initial energization criteria as described in Appendix A. The 2024 shoulder model (batteries charging) was used with all identified ERIS and NRIS Network Upgrades included. The results are summarized in Table 5.1-1. No constraints were found.

Table 5.1-1 – Inrush Calculations Using Shoulder Model

Generation Project	Transformer(s)		PSSE POI	Fault	Vmin Inrush (pu)	
	Number	Windings	Bus #	Current (A)	Raw	Multiplier
J1214	2	2	73270	16598	0.955	1.098
J1244	2	2	42447	2943	0.815	0.937
J1251	1	2	42527	5760	0.818	0.941

J1252	1	2	42527	5760	0.954	1.097
J1253	1	3	42535	11800	0.917	1.055
J1304	1	2	43044	21527	0.964	1.109
J1305	1	2	43054	5989	0.907	1.043
J1316	1	3	699409	21773	0.970	1.116
J1326	1	2	73270	16598	0.976	1.123
J1345	1	2	698222	7143	0.848	0.975
J1370	1	2	699713	3214	0.695	0.799
J1374	1	2	88554	13217	0.923	1.061
J1377	1	2	43774	12293	0.949	1.091
J1410	2	2	999907	16067	0.954	1.097
J1411	1	2	999907	16067	0.976	1.122
J1460	2	2	698095	14939	0.890	1.023
J1483	1	2	693863	9167	0.949	1.092

6.0 LOW SHORT CIRCUIT STRENGTH SCREENING ANALYSIS

ATC performed a low short circuit strength screening analysis for all inverter-based resources in the DPP 2019 Cycle 1 Phase 1 East (ATC) area. The methodology for the low short circuit strength screening analysis is detailed in Appendix B. PSCAD analysis will be performed as part of MISO DPP 2019 Cycle 1 Phase 2.

6.1 Low Short Circuit Strength Screening Results

The low short circuit strength screening analysis included classic Short Circuit Ratio (SCR) analysis with results shown in Tables 6.1-1, 6.1-2, and 6.1-3.

Table 6.1-1: SCR Analysis Results for System Intact

MISO Queue #	POI Bus #	Capacity (MW)	Short Circuit Capacity (MVA)	Short Circuit Ratio at POI
J1214/J1326	30000	375	9282.33	24.75
J1244	30001	38.4	347.93	9.06
J1251/J1252	30002	120	1414.02	11.78
J1253	30004	100	7153.00	71.53
J1304	41294	130	5231.03	40.24
J1305	41274	100	888.41	17.77
J1316/J878	699409	250	5324.68	21.30
J1345/J1003	698222	75	877.37	11.70
J1374/J855	88554	195	3092.97	15.86
J1377	30003	98.56	7497.78	76.07
J1410/J1411	694097	375	9816.47	26.18
J1460	698095	200	3623.22	18.12
J1483	693863	99	5311.83	53.65

Table 6.1-2: SCR Analysis Results for Worst N-1 Contingency

MISO Queue #	POI Bus #	Capacity (MW)	Short Circuit Capacity (MVA)	Short Circuit Ratio at POI
J1214/J1326	30000	375	2437.53	6.50
J1244	30001	38.4	146.62	3.82
J1251/J1252	30002	120	908.04	7.57
J1253	30004	100	3658.00	36.58
J1304	41294	130	2577.42	19.83
J1305	41274	100	488.44	4.88
J1316/J878	699409	250	3034.33	12.14
J1345/J1003	698222	75	572.77	7.64
J1374/J855	88554	195	604.00	3.10
J1377	30003	98.56	3107.68	31.53
J1410/J1411	694097	375	8043.27	21.45
J1460	698095	200	2478.35	12.39
J1483	693863	99	3135.50	31.67

Table 6.1-3: SCR Analysis Results for Worst N-1-1 Contingency

MISO Queue #	POI Bus #	Capacity (MW)	Short Circuit Capacity (MVA)	Short Circuit Ratio at POI
J1214/J1326	30000	375	2306.45	6.15
J1244	30001	38.4	65.32	1.70
J1251/J1252	30002	120	547.29	4.56
J1253	30004	100	1294.00	12.94
J1304	41294	130	1254.00	9.65
J1305	41274	100	232.94	2.33
J1316/J878	699409	250	1835.07	7.34
J1345/J1003	698222	75	156.71	2.09
J1374/J855	88554	195	399.89	2.05
J1377	30003	98.56	2802.63	28.44
J1410/J1411	694097	375	5179.48	13.81
J1460	698095	200	885.12	4.43
J1483	693863	99	945.26	9.55

It was determined that several groups of inverter-based resources were placed sufficiently close together to require Weighted Short Circuit Ratio (WSCR) analysis. The results for these WSCR analyses are shown in Tables 6.1-4, 6.1-5, and 6.1-6.

Table 6.1-4: WSCR Analysis Results for System Intact

Generator Group	Weighted SCR
Garden Wind, J849, J928, J1183, J1251, J1252	3.37
J855, J870, J871, J1000, J1374, J1483	4.20

Table 6.1-5: WSCR Analysis Results for Worst N-1 Contingency

Generator Group	Weighted SCR
Garden Wind, J849, J928, J1183, J1251, J1252	2.66
J855, J870, J871, J1000, J1374, J1483	3.35

Table 6.1-6: WSCR Analysis Results for Worst N-1-1 Contingency

Generator Group	Weighted SCR
Garden Wind, J849, J928, J1183, J1251, J1252	1.68
J855, J870, J871, J1000, J1374, J1483	1.84

Application of the ATC Low Short Circuit Strength Guideline determined that J1244, J1251, J1252, J1304, J1305, J1345, J1374, J1410, J1411, and J1460 will all require detailed PSCAD studies to be performed to assess stability. Table 6.1-7 below summarizes the rationale for the PSCAD analysis decision for each queue request.

Table 6.1-7: Required PSCAD Studies and Rationale

MISO Queue #	Low SCR	Low WSCR	East of Arnold Substation	Operational / Previous Study Experience	PSCAD Study Required?
J1214/J1326	No	No	No	No	No
J1244	Yes	No	No	Yes	Yes
J1251/J1252	No	No	Yes	No	Yes
J1253	No	No	No	No	No
J1304	No	No	No	Yes	Yes
J1305	No	No	No	Yes	Yes
J1316/J878	No	No	No	No	No
J1345/J1003	Yes	No	No	Yes	Yes
J1374/J855	Yes	No	No	Yes	Yes
J1377	No	No	No	No	No
J1410/J1411	No	No	No	Yes	Yes
J1460	No	No	No	Yes	Yes
J1483	No	No	No	Yes	Yes

7.0 AFFECTED SYSTEM ANALYSIS

Affected system analyses will be performed as part of MISO DPP 2019 Cycle 1 Phase 2.

8.0 DELIVERABILITY STUDY

Generator interconnection requests have to pass Generator Deliverability Study to be granted Network Resource Interconnection Services (NRIS). If the generator is determined as not fully deliverable, the customer can either choose to elect the amount of NRIS available without upgrades or build system upgrades that will make the generator fully deliverable. Generator Deliverability Study ensures that the Network Resources, on an aggregate basis, can meet the MISO aggregate load requirements during system peak condition without getting bottled up.

MISO Generator Deliverability Study methodology is described in MISO BPM-15.

8.1 Study Summary

The summary of MISO deliverability results based on the 2024 summer peak study model is shown in the following tables.

Table 8.1-1 below lists the deliverability results with ERIS upgrades included in the NRIS analysis. Minimum NR Deliverable is the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew. Maximum NR Deliverable is the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction.

Table 8.1-1 – NRIS Analysis Summary

MISO Queue #	Area	NR Tested (MW)	Minimum NR Deliverable (MW)	Maximum NR Deliverable (MW)
J1214	ATC	300	0	0
J1244	ATC	38.4	0	12.04
J1251	ATC	100	100	100
J1252	ATC	20	20	20
J1253	ATC	100	0	10.29
J1304	ATC	65	0	0
J1305	ATC	50	0	0
J1316	ATC	50	0	0
J1326	ATC	75	0	0
J1345	ATC	25	0	0
J1370	ATC	50	31.28	31.28
J1374	ATC	95	0	9.78
J1377	ATC	98.6	0	0
J1410	ATC	300	0	0
J1411	ATC	75	0	0
J1460	ATC	200	0	0
J1483	ATC	99	0	10.19

Table 8.1-2 below lists all of the NRIS constraints from the deliverability study and the identified NRIS Network Upgrades. Both ERIS Network Upgrades and NRIS Network Upgrades must be made for 100% NRIS, i.e. fully deliverable. Please note, if a NRIS Network Upgrade entirely or partially changes the scope of an ERIS Network Upgrade, only the cost difference between the NRIS upgrade and the ERIS upgrade will be eligible for NRIS Network Upgrade cost allocation. Detailed NRIS Network Upgrade cost allocation calculations are provided in Section 9.

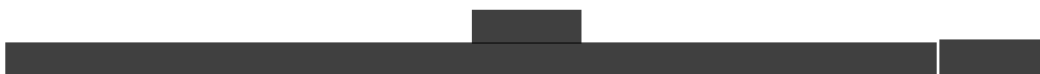


Table 8.1-2 – Network Upgrades Needed to Address MISO Identified NRIS Steady-State Injection Constraints

NRIS Thermal Constraints	Required NRIS Network Upgrades	Mitigation Type	Responsible Generators	NRIS Network Upgrade Cost Estimates (\$) [a]	ERIS Network Upgrade (Required for Same Constraint) Cost Estimate (\$) [b]	Previous Cycle ERIS/NRIS Network Upgrade (Identified for Same Constraint) Cost Estimate (\$) [c]	Cost Used for NRIS Network Upgrade Allocation (\$) [c] = [a] - [b]
43386 J1338 POI 345 345230 7MONTGMRY 345 1	J1338 POI - Montgomery 345 kV, Rebuild	NR	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	22,000,000	0	0	22,000,000
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1	Neevin – North Appleton 138 kV, New Line	NR	J1253, J1345, J1410, J1411	7,789,389	0	0	7,789,389
699512 UNVRSTY 138 699357 MUKWONGO 138 1	University – Mukwonago 138 kV (UNIG52), Rebuild	NR	J1304, J1305, J1460	31,814,057	0	0	31,814,057
698884 JEFRSN5 138 699375 CRWFSH R 138 1	Jefferson – Crawfish River 138 kV (6654), Rebuild	NR	J1214, J1326, J1377, J1410, J1411	3,147,000	0	0	3,147,000
699175 NFL 138 138 699677 AVIATION 138 1	North Fond Du Lac – Aviation 138 kV (G111), Rebuild	NR	J1345		0	0	0
699677 AVIATION 138 699663 PROGRESS AVE 138 1 699663 PROGRESS AVE 138 699673 EOD_BUS2 138 1	Aviation - Ellinwood 138 kV (X-50), Rebuild	NR	J1345	2,807,900	0	0	2,807,900
699749 NINEMILE 69.0 699475 KINCHOT 69.0 1 699475 KINCHOT 69.0 699776 TONE 69.0 1 699776 TONE 69.0 699725 RUDYRD 69.0 1 699725 RUDYRD 69.0 699751 PINERV 69.0 1 699749 NINEMILE 69.0 699751 PINERV 69.0 1	Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	NR	J1370	25,477,000	0	0	25,477,000
699115 WYV 138 138 699114 SPG 138 138 1 88704 J870J871 POI 138 699115 WYV 138 138 1	Spring Green – Highland 138 kV (X-17), Reconductor	NR	J1374, J1483	6,886,331	0	0	6,886,331
694022 FOXRIVER B1 345 699359 N APPLETON 345 1	Fox River – North Appleton 345 kV (NAPL121), Uprate	NR	J1316	292,629	0	0	292,629
699085 NLG 138 138 699360 NLK GV T 138 1	North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	NR	J1304, J1305, J1460	411,985	0	0	411,985
699360 NLK GV T 138 699267 BRLGTN1 138 1	North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	NR	J1304, J1305, J1460	4,309,017	0	3,963,594	345,423
698883 WHTWTR5 138 699516 BLUFFCRK 138 1 699516 BLUFFCRK 138 699488 SGR CK5 138 1	White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	NR	J1304, J1305, J1460	399,588	0	0	399,588
699086 ELK 138 138 699085 NLG 138 138 1	Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	NR	J1304, J1305, J1460	52,259	0	45,400	6,859
698995 MASS 69.0 698997 BRUCE CR 69.0 1 698997 BRUCE CR 69.0 698999 WATERSM 69.0 1	Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	NR	J1244	584,079	0	0	584,079
699114 SPG 138 138 699116 TRO 138 138 1	Spring Green – Troy 138 kV (X-18), Uprate	NR	J1374, J1483	444,134	0	0	444,134

699157 COL 345345 694069 SFL 345 B1345 1	Columbia – South Fond du Lac 345 kV (W-5), Uprate	NR	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1483	578,748	0	0	578,748
699340 JEFRSN4138 698884 JEFRSN5138 Z	Jefferson 138 kV, Replace Switches	NR	J1214, J1326, J1377	194,675	0	0	194,675
699375 CRWFSH R138 699283 CONCRD 4138 1	Crawfish River – Concord 138 kV (9043), Uprate	NR	J1214, J1326, J1377, J1410, J1411	2,084,375	0	902,919	1,181,456
89864 J986 POI138 699241 POE 138138 1	J986 POI – Port Edwards (X-11) 138 kV, Uprate	NR	J1345	4,652,851	0	584,677	4,068,174

8.2 Per Project Summary

8.2.1 J1214

J1214 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	300 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	2681.74	0	269.12	30.88	0.055	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	4,417,599	22,000,000	
698884 JEFRSN5 138 699375 CRWFSH R 138 1	1918.38	0	374.86	0	0.062	J1214, J1326, J1377, J1410, J1411	1,229,847	3,147,000	
699375 CRWFSH R 138 699283 CONCRD 4 138 1	1627.36	0	318	0	0.062	J1214, J1326, J1377, J1410, J1411	461,713	1,181,456	
699340 JEFRSN4 138 698884 JEFRSN5 138 Z	1368.22	0	446.56	0	0.063	J1214, J1326, J1377	127,045	194,675	
699157 COL 345 345 694069 SFL 345 B1 345 1	811.88	0	93.58	206.42	0.116	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483	133,401	578,748	
		300		300					

8.2.2 J1244

J1244 Deliverable (NRIS) Amount in 2023 Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	38.4 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
698995 MASS 69.0 698997 BRUCE CR 69.0 1	52.7	0	26.36	12.04	0.129	J1244	584,079	584,079	
698997 BRUCE CR 69.0 698999 WATERSM 69.0 1	10.2	28.2	5.1	33.3	0.128	J1244	Cost included in above constraint	Cost included in above constraint	
		38.4		38.4					

8.2.3 J1251

J1251 Deliverable (NRIS) Amount in 2023 Case: 100 MW (Conditional on ERIIS upgrades and case assumptions)	100 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes	
	100		100							

8.2.4 J1252

J1252 Deliverable (NRIS) Amount in 2023 Case: 20 MW (Conditional on ERIIS upgrades and case assumptions)	20MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
	20		20						

8.2.5 J1253

J1253 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	100 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1461.35	0	89.71	10.29	0.051	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	1,350,799	22,000,000	
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1	186.68	0	79.48	20.52	0.159	J1253, J1345, J1410, J1411	3,315,943	7,789,389	
		100		100					

8.2.6 J1304

J1304 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	65 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1371.76	0	58.31	6.69	0.054	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	934,999	22,000,000	
699157 COL 345 345 694069 SFL 345 B1 345 1	786.95	0	20.28	44.72	0.06	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483	14,932	578,748	
699512 UNVRSTY 138 699357 MUKWONGO 138 1	771.79	0	93.21	0	0.057	J1304, J1305, J1460	3,550,449	31,814,057	
699085 NLG 138 138 699360 NLK GV T 138 1	351.64	0	65.41	0	0.084	J1304, J1305, J1460	78,236	411,985	
699516 BLUFFCRK 138 699488 SGR CK5 138 1	129.48	0	11.05	53.95	0.05	J1304, J1305, J1460	34,085	399,588	
699360 NLK GV T 138 699267 BRLGTN1 138 1	85.78	0	12.88	52.12	0.088	J1304, J1305, J1460	51,883	345,423	
698883 WHTWTR5 138 699516 BLUFFCRK 138 1	58.43	6.57	4.98	60.02	0.05	J1304, J1305, J1460	Cost included in above constraint	Cost included in above constraint	
699086 ELK 138 138 699085 NLG 138 138 1	38.35	26.65	7.01	57.99	0.118	J1304, J1305, J1460	1,254	6,859	
		65		65					

8.2.7 J1305

J1305 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)							49.9 MW Requested											
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)							Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint			NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes	
43386	J1338	POI	345	345230	7MONTGMRY	345 1	1327.82	0	44.76	5.24	0.056	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483			741,399	22,000,000		
699512	UNVRSTY		138	699357	MUKWONGO	138 1	739.56	0	71.56	0	0.076	J1304, J1305, J1460			2,888,716	31,814,057		
699157	COL	345	345	694069	SFL 345 B1	345 1	727.79	0	15.57	34.43	0.065	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483			12,385	578,748		
699085	NLG	138	138	699360	NLK GV T	138 1	352.25	0	50.21	0	0.084	J1304, J1305, J1460			58,461	411,985		
699516	BLUFFCRK		138	699488	SGR CK5	138 1	116.25	0	8.48	41.42	0.056	J1304, J1305, J1460			29,170	399,588		
699360	NLK GV T		138	699267	BRLGTN1	138 1	83.8	0	9.89	40.01	0.09	J1304, J1305, J1460			40,725	345,423		
698883	WHTWTR5		138	699516	BLUFFCRK	138 1	52.46	0	3.83	46.07	0.056	J1304, J1305, J1460			Cost included in above constraint	Cost included in above constraint		
699086	ELK	138	138	699085	NLG 138	138 1	37.91	12.09	5.38	44.52	0.119	J1304, J1305, J1460			973	6,859		
								49.9		49.9								

8.2.8 J1316

J1316 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIS upgrades and case assumptions)										50 MW Requested									
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)										Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes	
694022 FOXRIVER B1 345 699359 N APPLETON 345 1										656.99	0	656.99	0	0.059	J1316	292,629	292,629		
											50		50						

8.2.9 J1326

J1326 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)							75 MW Requested											
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)							Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint			NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes	
43386	J1338	POI	345	345230	7MONTGMRY	345 1	1340.87	0	67.28	7.72	0.055	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483			1,104,399	22,000,000		
698884	JEFRSN5		138	699375	CRWFSH R	138 1	959.19	0	93.72	0	0.062	J1214, J1326, J1377, J1410, J1411			307,462	3,147,000		
699375	CRWFSH	R	138	699283	CONCRD 4	138 1	813.68	0	79.5	0	0.062	J1214, J1326, J1377, J1410, J1411			115,428	1,181,456		
699340	JEFRSN4		138	698884	JEFRSN5	138 Z	684.11	0	111.64	0	0.063	J1214, J1326, J1377			31,790	194,675		
699157	COL	345	345	694069	SFL 345 B1	345 1	405.94	0	23.4	51.6	0.116	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483			33,336	578,748		
								75		75								

8.2.10 J1345

J1345 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	25 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1413.37	0	22.43	2.57	0.053	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	349,799	22,000,000	
699175 NFL 138 138 699677 AVIATION 138 1	999.46	0	999.46	0	0.094	J1345	12,443,358	12,443,358	
699677 AVIATION 138 699663 PROGRESS AVE 138 1	759.49	0	759.49	0	0.094	J1345	2,807,900	2,807,900	
89864 J986 POI 138 699241 POE 138 1 C	546.16	0	546.16	0	0.051	J1345	4,068,174	4,068,174	
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1	362.31	0	19.87	5.13	0.082	J1253, J1345, J1410, J1411	427,637	7,789,389	
699663 PROGRESS AVE 138 699673 EOD_BUS2 138 1	203.3	0	203.3	0	0.094	J1345	Cost included in above constraint	Cost included in above constraint	
		25		25					

8.2.11 J1370

J1370 Deliverable (NRIS) Amount in * Case: 31.28 MW (Conditional on ERIIS upgrades and case assumptions)	50 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
699749 NINEMILE 69.0 699475 KINCHOT 69.0 1	18.72	31.28	18.72	31.28	0.72	J1370	25,477,000	25,477,000	
699749 NINEMILE 69.0 699751 PINERV 69.0 1	14.85	35.15	14.85	35.15	0.732	J1370	Cost included in above constraint	Cost included in above constraint	
699475 KINCHOT 69.0 699776 TONE 69.0 1	7.43	42.52	7.43	42.52	0.72	J1370	Cost included in above constraint	Cost included in above constraint	
699776 TONE 69.0 699725 RUDYRD 69.0 1	6.29	43.71	6.29	43.71	0.72	J1370	Cost included in above constraint	Cost included in above constraint	
699725 RUDYRD 69.0 699751 PINERV 69.0 1	4.58	45.42	4.58	45.42	0.72	J1370	Cost included in above constraint	Cost included in above constraint	
		50		50					

8.2.12 J1374

J1374 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	95 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1173.01	0	85.22	9.78	0.063	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	1,597,199	22,000,000	
699157 COL 345 345 694069 SFL 345 B1 345 1	484.75	0	29.64	65.36	0.054	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483	35,362	578,748	
88704 J870J871 POI 138 699115 WYV 138 138 1	143.39	0	84.58	10.42	0.158	J1374, J1483	4,062,247	6,886,331	
699115 WYV 138 138 699114 SPG 138 138 1	123.06	0	72.59	22.41	0.158	J1374, J1483	Cost included in above constraint	Cost included in above constraint	
699114 SPG 138 138 699116 TR0 138 138 1	30.44	64.56	17.89	77.11	0.121	J1374, J1483	260,929	444,134	
		95		95					

8.2.13 J1377

J1377 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	98.6 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1344.56	0	88.45	10.15	0.0552	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	1,445,399	22,000,000	
698884 JEFRSN5 138 699375 CRWFSH R 138 1	1128.21	0	123.2	0	0.0527	J1214, J1326, J1377, J1410, J1411	343,652	3,147,000	
699375 CRWFSH R 138 699283 CONCRD 4 138 1	957.06	0	104.51	0	0.0527	J1214, J1326, J1377, J1410, J1411	129,015	1,181,456	
699340 JEFRSN4 138 698884 JEFRSN5 138 Z	797.49	0	146.77	0	0.0538	J1214, J1326, J1377	35,840	194,675	
699157 COL 345 345 694069 SFL 345 B1 345 1	475.62	0	30.76	67.84	0.0992	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483	37,445	578,748	
		98.6		98.6					

8.2.14 J1410

Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	2666.36	0	269.12	30.88	0.056	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	4,439,599	22,000,000	
698884 JEFRSN5 138 699375 CRWFSH R 138 1	2329.58	0	374.86	0	0.051	J1214, J1326, J1377, J1410, J1411	1,012,705	3,147,000	
699375 CRWFSH R 138 699283 CONCRD 4 138 1	1976.18	0	692.86	0	0.051	J1214, J1326, J1377, J1410, J1411	380,193	1,181,456	
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1	1147.78	0	238.46	61.54	0.052	J1253, J1345, J1410, J1411	3,236,491	7,789,389	
699157 COL 345 345 694069 SFL 345 B1 345 1	494.76	0	93.58	206.42	0.191	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483	218,940	578,748	
		300		300					

8.2.15 J1411

J1411 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	75 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1333.18	0	67.28	7.72	0.056	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	1,110,999	22,000,000	
698884 JEFRSN5 138 699375 CRWFSH R 138 1	1164.79	0	93.72	0	0.051	J1214, J1326, J1377, J1410, J1411	253,334	3,147,000	
699375 CRWFSH R 138 699283 CONCRD 4 138 1	988.09	0	79.5	0	0.051	J1214, J1326, J1377, J1410, J1411	95,107	1,181,456	
699551 NEEVIN-WEC 138 699525 BD MRT2 138 1	573.89	0	59.61	15.39	0.052	J1253, J1345, J1410, J1411	809,318	7,789,389	
699157 COL 345 345 694069 SFL 345 B1 345 1	247.38	0	23.4	51.6	0.191	J1214, J1304, J1305, J1326, J1374, J1377, J1410, J1411, J1460, J1483	54,750	578,748	
		75		75					

8.2.16 J1460

J1460 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIIS upgrades and case assumptions)	200 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project (\$)	Total NRIS Cost of Upgrade (\$)	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	2793.06	0	179.42	20.58	0.053	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	2,826,999	22,000,000	
699512 UNVRSTY 138 699357 MUKWONGO 138 1	733.08	0	286.8	0	0.121	J1304, J1305, J1460	25,374,892	31,814,057	
699085 NLG 138 138 699360 NLK GV T 138 1	599.5	0	201.26	0	0.098	J1304, J1305, J1460	275,288	411,985	
699360 NLK GV T 138 699267 BRLGTN1 138 1	108.32	91.68	39.64	160.36	0.139	J1304, J1305, J1460	252,815	345,423	
699516 BLUFFCRK 138 699488 SGR CK5 138 1	80.76	119.24	33.98	166.02	0.161	J1304, J1305, J1460	336,333	399,588	
699086 ELK 138 138 699085 NLG 138 138 1	63.86	136.14	21.56	178.44	0.141	J1304, J1305, J1460	4,632	6,859	
698883 WHTWTR5 138 699516 BLUFFCRK 138 1	36.44	163.56	15.34	184.66	0.161	J1304, J1305, J1460	Cost included in above constraint	Cost included in above constraint	
		200		200					

8.2.17 J1483

J1483 Deliverable (NRIS) Amount in * Case: 0 MW (Conditional on ERIS upgrades and case assumptions)	99 MW Requested								
Next Constraint for Higher NRIS Level (cumulative) (i.e. All upgrades must be made for 100% NRIS)	Single Deduction	Minimum Level of Service Attainable (MW) *This would be the amount of service to be granted if none of the projects contributing to this constraint, reduced service levels or withdrew	Shared Deduction	Maximum level of service Attainable (MW) *This would be the service level attainable if all projects contributing to the constraint, reduced their requested NR by the shared deduction	Max DF	Projects Associated With NRIS Constraint	NRIS Cost Allocated to Project	Total NRIS Cost of Upgrade	Notes
43386 J1338 POI 345 345230 7MONTGMRY 345 1	1162.38	0	88.81	10.19	0.064	J1214, J1253, J1304, J1305, J1326, J1345, J1374, J1377, J1410, J1411, J1460, J1483	1,680,799	22,000,000	
699157 COL 345 345 694069 SFL 345 B1 345 1	468.19	0	30.88	68.12	0.101	J1460, J1483	38,197	578,748	
88704 J870J871 POI 138 699115 WYV 138 138 1	214.89	0	88.14	10.86	0.105	J1374, J1483	2,824,084	6,886,331	
699115 WYV 138 138 699114 SPG 138 138 1	184.42	0	75.64	23.36	0.105	J1374, J1483	Cost included in above constraint	Cost included in above constraint	
699114 SPG 138 138 699116 TR0 138 138 1	45.21	53.79	18.64	80.36	0.081	J1374, J1483	183,205	444,134	
		99		99					

8.3 Network Upgrade Alternatives Considered

All NRIS network upgrades identified in Table 8.1-2 except the Neevin – North Appleton 138 kV new line project are direct upgrades of the constraint facilities to ATC and Ameren design standards and considered as least-cost solutions at this point. Therefore, no other alternatives were examined. As an alternative to the Neevin – North Appleton 138 kV new line project, a new 138 kV line between Neevin to Butte Des Morts was considered but dismissed due to a higher cost.

9.0 COST ALLOCATION AND TRANSMISSION OWNER SELF-FUND ELECTION

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the System Impact Study report date. The cost estimate for each network upgrade was provided by the corresponding transmission owning company.

Per MISO tariff, Transmission Owners must provide a non-binding indication of whether or not they will self-fund any Network Upgrades identified in the System Impact Study analysis. The Self-Funding elections are shown in sections 9.1, 9.2, and 9.3.

9.1 Interconnection Facilities Proposed for DPP 2019 Cycle 1 Phase 1 Projects

For DPP 2019 Cycle 1 Phase 1 East (ATC) group, the total costs of interconnection facilities are summarized in Table 9.1-1.

Table 9.1-1 – Interconnection Facilities for DPP 2019 Cycle 1 Phase 1 Projects

MISO Queue #	Interconnection Facilities (\$)	
	TO Network Upgrades	Self-Fund
[a]	[h]	[i]
J1214	859,833	No
J1244	6,385,560	No
J1251	3,956,000	No
J1252	3,956,000	No
J1253	13,498,520	No
J1304	7,836,034	No
J1305	7,998,040	No
J1316	0	NA
J1326	859,833	No
J1345	0	NA
J1370	0	NA
J1374	767,975	No
J1377	11,966,628	No
J1410	5,336,386	No
J1411	5,336,386	No

J1460	4,881,028	No
J1483	1,447,638	No

9.2 ERIS Network Upgrades Proposed for DPP 2019 Cycle 1 Phase 1 Projects

Network upgrades for Energy Resource Interconnection Service (ERIS) were identified in the ERIS analysis. The ERIS network upgrades include thermal network upgrades and voltage support network upgrades identified in the steady-state analysis. For DPP 2019 Cycle 1 Phase 1 East (ATC) group, the total costs of ERIS network upgrades for the 2024 scenario are summarized in Tables 9.2-1, 9.2-2, 9.2-3, 9.2-4, 9.2-5, 9.2-6 and 9.2-7.

Table 9.2-1 – ERIS Network Upgrades Identified in Steady State Analysis

Steady-State Injection Constraint	Facility Owner	Network Upgrade	ISD (For Cost Estimate Only)	Cost (\$) ^{1,2}	Self-Fund
Dafter – Nine Mile 69 kV line	ATC	Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate	3/01/2021	325,098	No
J1244 POI – Portage Tap 69 kV line	ATC	Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP PrjId#4727	12/01/2024	0	NA
Portage Tap – Atlantic 69 kV line	ATC				
J1244 POI – Winona 69 kV line	ATC				
Highland – Eden 138 kV line	ATC	Highland – Eden (X-147) 138 kV, Rebuild	10/30/2023	3,518,467	No

¹ All Network Upgrades were estimated on the earliest ISD dollars of responsible generator.

² ATC Network Upgrades included a 10% contingency.

Table 9.2-2 – ERIS Network Upgrades in Dynamic Stability Analysis

To be determined in MISO DPP 2019 Cycle 1 Phase 2

Table 9.2-3 – ERIS Network Upgrades in Short Circuit Analysis

To be determined in MISO DPP 2019 Cycle 1 Phase 2

Table 9.2-4 – ERIS Affected System Network Upgrades

To be determined in MISO DPP 2019 Cycle 1 Phase 2

Table 9.2-5 – ERIS Network Upgrades Identified in PSCAD Analysis

To be determined in MISO DPP 2019 Cycle 1 Phase 2

Table 9.2-6 – ERIS PJM Affected System Network Upgrades

To be determined in MISO DPP 2019 Cycle 1 Phase 2

Table 9.2-7 – ERIS Shared Network Upgrades

Constraint	Facility Owner	Network Upgrade	Cost (\$)
None	-	-	-

9.3 NRIS Network Upgrades Proposed for DPP 2019 Cycle 1 Phase 1 Projects

Network upgrades for Network Resource Interconnection Service (NRIS) were identified in the MISO's deliverability analysis and listed in the Table 9.3-1 below.

Table 9.3-1 – NRIS Network Upgrades Identified

Network Upgrade	Facility Owner	ISD (For Cost Estimate Only) ¹	Cost Used for NRIS Cost Allocation (\$) ²	Self-Fund
J1338 POI - Montgomery 345 kV, Rebuild	AMMO	8/01/2021	22,000,000 ³	Yes
Neevin – North Appleton 138 kV, New Line	ATC	10/01/2021	7,789,389	No
University – Mukwonago 138 kV (UNIG52), Rebuild	ATC	12/01/2021	31,814,057	No
Jefferson – Crawfish River 138 kV (6654), Rebuild	ATC	7/15/2022	3,147,000	No
North Fond Du Lac – Aviation 138 kV (G111), Rebuild	ATC	10/01/2021	12,443,358	No
Aviation - Ellinwood 138 kV (X-50), Rebuild	ATC	10/01/2021	2,807,900	No
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	ATC	3/01/2021	25,477,000	No
Spring Green – Highland 138 kV (X-17), Reconductor	ATC	8/01/2021	6,886,331	No
Fox River – North Appleton 345 kV (NAPL121), Uprate	ATC	10/30/2022	292,629	No
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	ATC	12/01/2021	411,985	No
North Lake Geneva Tap – Burlington 138 kV (6541), Reconductor	ATC	12/01/2021	345,423	No
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	ATC	12/01/2021	399,588	No
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	ATC	12/01/2021	6,859	No
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	ATC	9/15/2021	584,079	No
Spring Green – Troy 138 kV (X-18), Uprate	ATC	8/01/2021	444,134	No
Columbia – South Fond du Lac 345 kV (W-5), Uprate	ATC	8/01/2021	578,748	No
Jefferson 138 kV, Replace Switches	ATC	7/15/2022	194,675	No
Crawfish River – Concord 138 kV (9043), Uprate	ATC	7/15/2022	1,181,456	No
J986 POI – Port Edwards (X-11) 138 kV, Uprate	ATC	10/01/2021	4,068,174	No

¹ All Network Upgrades were estimated on the earliest ISD dollars of responsible generators.

² ATC Network Upgrades included a 10% contingency.

³ This constraint is yet to be evaluated in the DPP 2019 Cycle 1 Phase 1 Central and West area studies and will be updated after those studies are completed.

9.4 Cost Allocation Methodology for Thermal Network Upgrades

The costs of Network Upgrades (NU) for a set of generation projects (one or more subgroups or entire group with identified NU) are based off the MW impact of the worst case scenario for each specific generator project. Basically, whatever the

highest MW impact (increasing flow) is for that particular generator where the constraint is identified and requires NU is how it should be calculated.

Constraints which are mitigated by one or a subset of NU are identified. The highest MW contribution on these constraints from each generating facility is calculated in the MISO DPP study models without any Network Upgrades. Then the cost of each NU is allocated based on the pro rata share of the MW contribution from each generating facility on the constraints mitigated or partly mitigated by this NU. The methodology to determine the cost allocation of NU is:

$$\text{Project A cost portion of NU} = \text{Cost of NU} \times \left(\frac{\text{Max}(\text{Proj. A MW contribution on constraint})}{\sum_i \text{Max}(\text{Proj. i MW contribution on constraint})} \right)$$

9.5 Cost Estimating and Allocation Methodology for Short Circuit Upgrades

For each ATC breaker shown to be overdutied a new breaker will be scoped and the cost of that upgrade will be assigned to generators. No breaker duty analysis is performed for Affected System equipment. Appendix N provides short circuit data for all substation buses 69 kV and above included in ATC protection CAPE models that show a 1% increase in SLG or 3PG fault current. Any non-ATC equipment requiring a change would be considered Affected System.

For each ATC substation that shows a bus having fault current (SLG) increased by 10% or more, ATC will perform detailed grounding grid analysis if in-house grounding grid modeling is available at the time of the study. For the substations that pass the detailed grounding analysis (i.e. meet the safety criteria), they will be removed from the cost allocation process. For the remaining substations, costs are assigned for ground grid upgrades at that substation.

Only one cost estimate is scoped per substation regardless of the number of buses at that substation that show a 10% or greater increase in fault current. The largest fault current value at a substation in the After Case will determine the ground grid upgrade costs. If the largest fault current is above 20 kA, the upgrades are assigned a cost estimate of \$550,000. If the largest fault current is at or below 20 kA, the upgrades are assigned a cost estimate of \$220,000 for ground grid upgrades only. These are placeholder costs until Facility Studies are performed. Based on those results, costs may be adjusted.

The ground grid study methodology (10% or greater increase in SLG fault current) is applied to both ATC and non-ATC facilities modeled in ATC Protection models. For non-ATC facilities, it is interconnection customer's responsibility to work with facility owners to develop costs, and implementation of the mitigation projects (if needed).

Once breaker replacement costs and substation ground grid upgrade costs are determined, they are allocated proportionally to study generators that have a greater than 3% of the total of all current queue study generator contributing fault currents under the single line-ground short circuit fault simulation at the identified substation.

9.6 Cost Allocation Tables

The cost allocation of Network Upgrades for the study group reflects responsibilities for mitigating system impacts based on Interconnection Customer-elected level of Network Resource Interconnection service as of the issued date of the System Impact Study report.

Assuming all generating facilities in the DPP 2019 Cycle 1 Phase 1 (East ATC) group advance, Table 9.6-1, Table 9.6-2, Table 9.6-3, Table 9.6-4, and Table 9.6-5 show how the costs for ERIS steady-state Network Upgrades, short-circuit Network Upgrades, EMT stability Network Upgrades, PJM affected system Network Upgrades and NRIS Network Upgrades allocated to responsible generating facilities.

Table 9.6-1 – ERIS Steady-State Thermal Network Upgrade Costs Allocated to Each Generation Project

Required ERIS Network Upgrades	Worst MW Impact, % Cost Allocation			Total MW Impact	Cost Allocation (\$)		
	J1244	J1370	J1483		J1244	J1370	J1483
Dafter – Nine Mile (ESE_6957) 69 kV, Bifurcate	-	50, 100	-	50.0	-	325,098	-
Winona - Atlantic (Winona69) 69kV, Reconductor; MTEP PrjId#4727	38.4, 100	-	-	38.4	0	-	-
Highland – Eden (X-147) 138 kV, Rebuild	-	-	12.2, 100	12.2	-	-	3,518,467
Total ERIS Steady State Thermal Network Upgrade Cost (\$) Allocated to Each Generator					0	325,098	3,518,467

Table 9.6-2 – ERIS Short-Circuit Network Upgrade Costs Allocated to Each Generation Project

To be determined in MISO DPP 2019 Cycle 1 Phase 2.

Table 9.6-3 – ERIS EMT Stability Network Upgrade Costs Allocated to Each Generation Project

To be determined in MISO DPP 2019 Cycle 1 Phase 2.

Table 9.6-4 – PJM Affected System Costs Allocated to Each Generation Project

To be determined in MISO DPP 2019 Cycle 1 Phase 2.

Table 9.6-5 – NRIS Network Upgrade Costs Allocated to Each Generation Project, Part 1

Required NRIS Network Upgrades	Worst MW Impact and % Cost Allocation														
	J1214	J1244	J1253	J1304	J1305	J1316	J1326	J1345	J1370	J1374	J1377	J1410	J1411	J1460	J1483
J1338 POI - Montgomery 345 kV, Rebuild ¹	16.62, 20.08	-	5.08, 6.14	3.52, 4.25	2.79, 3.37	-	4.16, 5.02	1.31, 1.59	-	6.01, 7.26	5.44, 6.57	16.71, 20.18	4.18, 5.05	10.64, 12.85	6.33, 7.64
Neevin – North Appleton 138 kV, New Line	-	-	15.90, 42.57	-	-	-	-	2.05, 5.49	-	-	-	15.52, 41.55	3.88, 10.39	-	-
University – Mukwonago 138 kV (UNIG52), Rebuild	-	-	-	4.63, 11.16	3.77, 9.08	-	-	-	-	-	-	-	-	33.10, 79.76	-
Jefferson – Crawfish River 138 kV (6654), Rebuild	18.60, 39.08	-	-	-	-	-	4.65, 9.77	-	-	-	5.20, 10.92	15.32, 32.18	3.83, 8.05	-	-
North Fond Du Lac – Aviation 138 kV (G111) , Rebuild	-	-	-	-	-	-	-	2.34, 100.00	-	-	-	-	-	-	-
Aviation - Ellinwood 138 kV (X-50), Rebuild	-	-	-	-	-	-	-	2.34, 100.00	-	-	-	-	-	-	-
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	-	-	-	-	-	-	-	-	35.98, 100.00	-	-	-	-	-	-
Spring Green – Highland 138 kV (X-17), Rebuild	-	-	-	-	-	-	-	-	-	14.99, 58.99	-	-	-	-	10.42, 41.01
Fox River – North Appleton 345 kV (NAPL121), Uprate	-	-	-	-	-	2.97, 100.00	-	-	-	-	-	-	-	-	-
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	-	-	-	5.58, 18.99	4.17, 14.19	-	-	-	-	-	-	-	-	19.64, 66.82	-
North Lake Geneva Tap – Burlington 138 kV (6541), Uprate	-	-	-	5.72, 15.02	4.49, 11.79	-	-	-	-	-	-	-	-	27.88, 73.19	-
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	-	-	-	3.27, 8.53	2.80, 7.30	-	-	-	-	-	-	-	-	32.28, 84.17	-
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	-	-	-	7.65, 18.28	5.94, 14.19	-	-	-	-	-	-	-	-	28.26, 67.53	-
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	-	4.96, 100.00	-	-	-	-	-	-	-	-	-	-	-	-	-
Spring Green – Troy 138 kV (X-18), Uprate	-	-	-	-	-	-	-	-	-	11.48, 58.75	-	-	-	-	8.06, 41.25
Columbia – South Fond du Lac 345 kV (W-5), Uprate	34.88, 23.05	-	-	-	-	-	3.90, 2.58	-	-	-	3.24, 2.14	57.24, 37.83	14.31, 6.60	-	-
Jefferson 138 kV, Replace Switches	18.82, 65.26	-	-	-	-	-	4.71, 16.33	-	-	-	5.31, 18.41	-	-	-	-
Crawfish River – Concord 138 kV (9043), Uprate	18.60, 39.08	-	-	-	-	-	4.65, 9.77	-	-	-	5.20, 10.92	15.32, 32.18	3.83, 8.05	-	-
J986 POI – Port Edwards (X-11) 138 kV, Uprate	-	-	-	-	-	-	-	1.28, 100.00	-	-	-	-	-	-	-

¹ This constraint is yet to be evaluated in the DPP 2019 Cycle 1 Phase 1 Central and West area studies and will be updated after those studies are completed.

Table 9.6-5 – NRIS Network Upgrade Costs Allocated to Each Generation Project, Part 2

Required NRIS Network Upgrades	Cost Allocation (\$)														
	J1214	J1244	J1253	J1304	J1305	J1316	J1326	J1345	J1370	J1374	J1377	J1410	J1411	J1460	J1483
J1338 POI - Montgomery 345 kV, Rebuild ¹	4,417,599	-	1,350,799	934,999	741,399	-	1,104,399	349,799	-	1,597,199	1,445,399	4,439,599	1,110,999	2,826,999	1,680,799
Neevin – North Appleton 138 kV, New Line	-	-	3,315,943	-	-	-	-	427,637	-	-	-	3,236,491	809,318	-	-
University – Mukwonago 138 kV (UNIG52), Rebuild	-	-	-	3,550,449	2,888,716	-	-	-	-	-	-	-	-	25,374,892	-
Jefferson – Crawfish River 138 kV (6654), Rebuild	1,229,847	-	-	-	-	-	307,462	-	-	-	343,652	1,012,705	253,334	-	-
North Fond Du Lac – Aviation 138 kV (G111) , Rebuild	-	-	-	-	-	-	-	12,443,358	-	-	-	-	-	-	-
Aviation - Ellinwood 138 kV (X-50), Rebuild	-	-	-	-	-	-	-	2,807,900	-	-	-	-	-	-	-
Nine Mile – Pine River 69 kV (6923/6921), Double Circuit Rebuild	-	-	-	-	-	-	-	-	25,477,000	-	-	-	-	-	-
Spring Green – Highland 138 kV (X-17), Rebuild	-	-	-	-	-	-	-	-	-	4,062,247	-	-	-	-	2,824,084
Fox River – North Appleton 345 kV (NAPL121), Uprate	-	-	-	-	-	292,629	-	-	-	-	-	-	-	-	-
North Lake Geneva – North Lake Geneva Tap 138 kV (6541), Uprate	-	-	-	78,236	58,461	-	-	-	-	-	-	-	-	275,288	-
North Lake Geneva Tap – Burlington 138 kV (6541), Uprate	-	-	-	51,883	40,725	-	-	-	-	-	-	-	-	252,815	-
White Water – Bluff Creek – Sugar Creek 138 kV (WHIG53), Uprate	-	-	-	34,085	29,170	-	-	-	-	-	-	-	-	336,333	-
Elkhorn – North Lake Geneva 138 kV (X-55), Uprate	-	-	-	1,254	973	-	-	-	-	-	-	-	-	4,632	-
Mass – Bruce Crossing – Watersmeet 69 kV (6530), Uprate	-	584,079	-	-	-	-	-	-	-	-	-	-	-	-	-
Spring Green – Troy 138 kV (X-18), Uprate	-	-	-	-	-	-	-	-	-	260,929	-	-	-	-	183,205
Columbia – South Fond du Lac 345 kV (W-5), Uprate	133,401	-	-	14,932	12,385	-	33,336	-	-	35,362	37,445	218,940	54,750	-	38,197
Jefferson 138 kV, Replace Switches	127,045	-	-	-	-	-	31,790	-	-	-	35,840	-	-	-	-
Crawfish River – Concord 138 kV (9043), Uprate	461,713	-	-	-	-	-	115,428	-	-	-	129,015	380,193	95,107	-	-
J986 POI – Port Edwards (X-11) 138 kV, Uprate	-	-	-	-	-	-	-	4,068,174	-	-	-	-	-	-	-
Total NRIS Steady State Thermal Network Upgrade Cost (\$) Allocated to Each Generator	6,369,605	584,079	4,666,742	4,665,837	3,771,830	292,629	1,592,415	20,096,868	25,477,000	5,955,736	1,991,351	9,287,928	2,323,507	29,070,959	4,726,286

¹ This constraint is yet to be evaluated in the DPP 2019 Cycle 1 Phase 1 Central and West area studies and will be updated after those studies are completed.



10.0 AVAILABLE APPENDIX DOCUMENTS (NOT ATTACHED)

Appendix A – Study Criteria, Methodology, and Assumptions

Appendix B – ATC Planning Criteria and Generation Facility Interconnection Guide

Appendix C – Interconnection Facility Project Diagrams and Modeling Details

Appendix D – Network Upgrade Project Diagrams

Note: Project Diagrams were not developed for line uprate projects.

Appendix E – Steady State Power Flow Results

Appendix I – MISO Deliverability Study Results

Appendix J – Assessed System Performance Reference